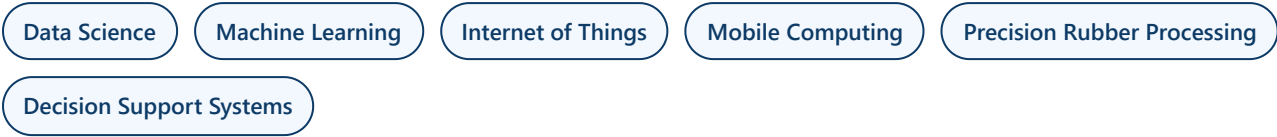


Intelligent Framework for Precision Rubber Processing and Supply Chain Optimization

Complete Research Framework, System Architecture, Data Pipeline and Implementation Plan



- 1 Smartphone-Based ML Decision Support for Rubber Latex Coagulation Management
- 2 Village-Level Weather & Tapping Advisory System
- 3 Mobile Acoustic “Thud-Test” App for Rubber Sheet Quality Assessment
- 4 Smart Smokehouse Drying-Time Prediction and Fire-Risk Monitoring System (primary component)

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Important Notice

Notice on Data Authenticity

This document is a complete research-plan and reference document for a university final-year research project. It describes the overall research topic, framework, four research components, datasets, machine-learning models, IoT architecture, mobile applications, workflows, system architecture, and related methodology in full technical detail.

Wherever this document includes sample datasets, illustrative figures, mock-up screens, or example numeric values, they are clearly labelled as **illustrative / synthetic** and are used only to demonstrate dataset structure, system behaviour, or user-interface design. They are **not** experimental results.

Where a specific performance value (such as an R^2 , RMSE, or MAE figure) is stated without an illustrative label, it is explicitly identified as a previously referenced/available result associated with the corresponding research line, and is not a new result produced by this document.

Where a value has not yet been experimentally established, this document states **“To be determined through experimentation”** rather than inventing a plausible-looking number. Final results for every component must be generated only from real, field-collected data.

1 Introduction

1.1 Research Topic and Domain

The research is titled **“Intelligent Framework for Precision Rubber Processing and Supply Chain Optimization.”** It is positioned at the intersection of several academic and applied domains, including data science, machine learning, the Internet of Things (IoT), mobile computing, precision rubber processing, agricultural technology, the smallholder rubber sector, and decision support systems.

The research focuses on developing an intelligent framework that improves rubber processing and related decision-making by combining IoT sensing, machine learning, mobile computing, and data-driven analytical approaches. Rather than treating rubber processing as a single isolated activity, the framework addresses several interconnected decision points that a smallholder rubber processor or farmer encounters, from the moment latex is collected to the point at which processed rubber sheets are dried and prepared for the supply chain.

1.2 Research Vision and Motivation

Traditional rubber processing depends heavily on manual measurement, farmer experience, fixed recommendations, manual quality assessment, uncertain weather interpretation, manual smokehouse monitoring, and limited real-time decision support. These practices are functional, but they are also subjective, inconsistent between operators, and difficult to scale or standardise across a supply chain that involves many smallholder producers.

The proposed research develops an integrated intelligent framework that uses machine learning, IoT, smartphone and mobile computing, weather intelligence, acoustic signal processing, environmental sensing, cloud/backend technologies, data analytics, and decision-support mechanisms. The overall vision is to transform selected rubber-processing activities from manual, experience-based processes into data-driven, intelligent decision support that remains accessible to farmers through ordinary mobile devices.

1.3 Research Problem

Smallholder rubber processing currently relies on manual Metrolac readings and reference charts for latex quality estimation, informal judgement for tapping and weather decisions, subjective tactile or visual inspection for sheet quality, and manual, infrequent observation of smokehouse conditions during drying. Each of these steps is a potential source of inconsistency, inefficiency, and, in the case of smokehouse operation, safety risk from unmonitored heat build-up. There is presently no single framework that connects these decision points using a shared, data-driven approach that is deployable on the mobile devices already used by farmers.

1.4 Research Objectives

- To design an integrated intelligent framework connecting four complementary rubber-processing decision-support components.
- To develop a smartphone-based decision-support approach for temperature-corrected latex coagulation management (Component 1).
- To design a village-level weather and tapping advisory system that supports tap / do-not-tap decisions (Component 2).
- To investigate a mobile acoustic “thud-test” approach for rubber-sheet quality assessment (Component 3).
- To design and prototype an IoT-enabled smart smokehouse system that predicts layer-wise remaining drying time and monitors fire risk (Component 4, the primary component of this research).
- To define a correlation analysis and data-preprocessing methodology suitable for the smokehouse drying dataset.

- To design a backend and mobile-application architecture capable of delivering the above decision support to farmers in real time.

1.5 Research Gap

The proposed research aims to address several gaps that are observable in current smallholder rubber-processing practice: fragmented decision support across the different stages of processing; limited integration of IoT and machine learning specifically for smokehouse drying; limited layer-wise (rather than whole-smokehouse) environmental monitoring; limited dynamic, data-driven prediction of remaining drying time; limited mobile-based processing decision support for smallholder farmers; continued reliance on manual environmental monitoring; and limited data-driven safety monitoring for fire risk during drying.

Note on novelty claims

Wording throughout this document intentionally uses phrases such as *"the proposed research aims to address..."* rather than absolute claims such as *"no previous system has ever..."*. Novelty is described as the proposed integration and application of established techniques to this domain, not as a claim that each individual technology is new.

2 Overall Intelligent Framework

2.1 Input–Process–Output Model

At the highest level, the framework can be described using a simple Input–Process–Output (IPO) model. The **inputs** comprise agricultural and processing data collected across the four components: Metrolac reading, latex temperature, latex volume, weather data (rainfall, humidity, temperature, wet hours), historical latex density and DRC, tapping frequency, rubber-sheet acoustic signals, smokehouse temperature and humidity, sheet-loading information, and historical drying records.

The **process** stage covers data collection, IoT sensing, smartphone data capture, cloud/local storage, data preprocessing, feature extraction and engineering, machine learning, signal processing, prediction, risk analysis, decision-support generation, and mobile notification. The **outputs** are the component-specific recommendations described in Table 1 and detailed in Chapters 3–6.

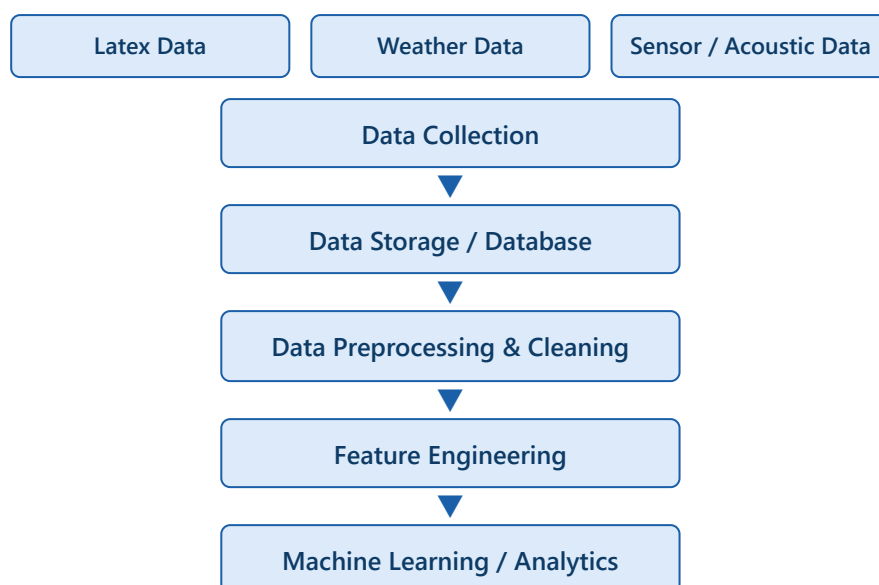


Figure 2. Input–Process–Output model of the overall intelligent framework.

2.2 Overall Framework Architecture

Figure 1 presents the complete architecture of the intelligent framework. Data collected from latex measurement, weather sources, and IoT/acoustic sensors flows through a common data-collection and storage layer, is cleaned and engineered into model-ready features, and is then processed by machine-learning and analytics modules specific to each of the four components. The outputs of all four components are unified at a decision-support layer and delivered to the farmer through a single mobile application.

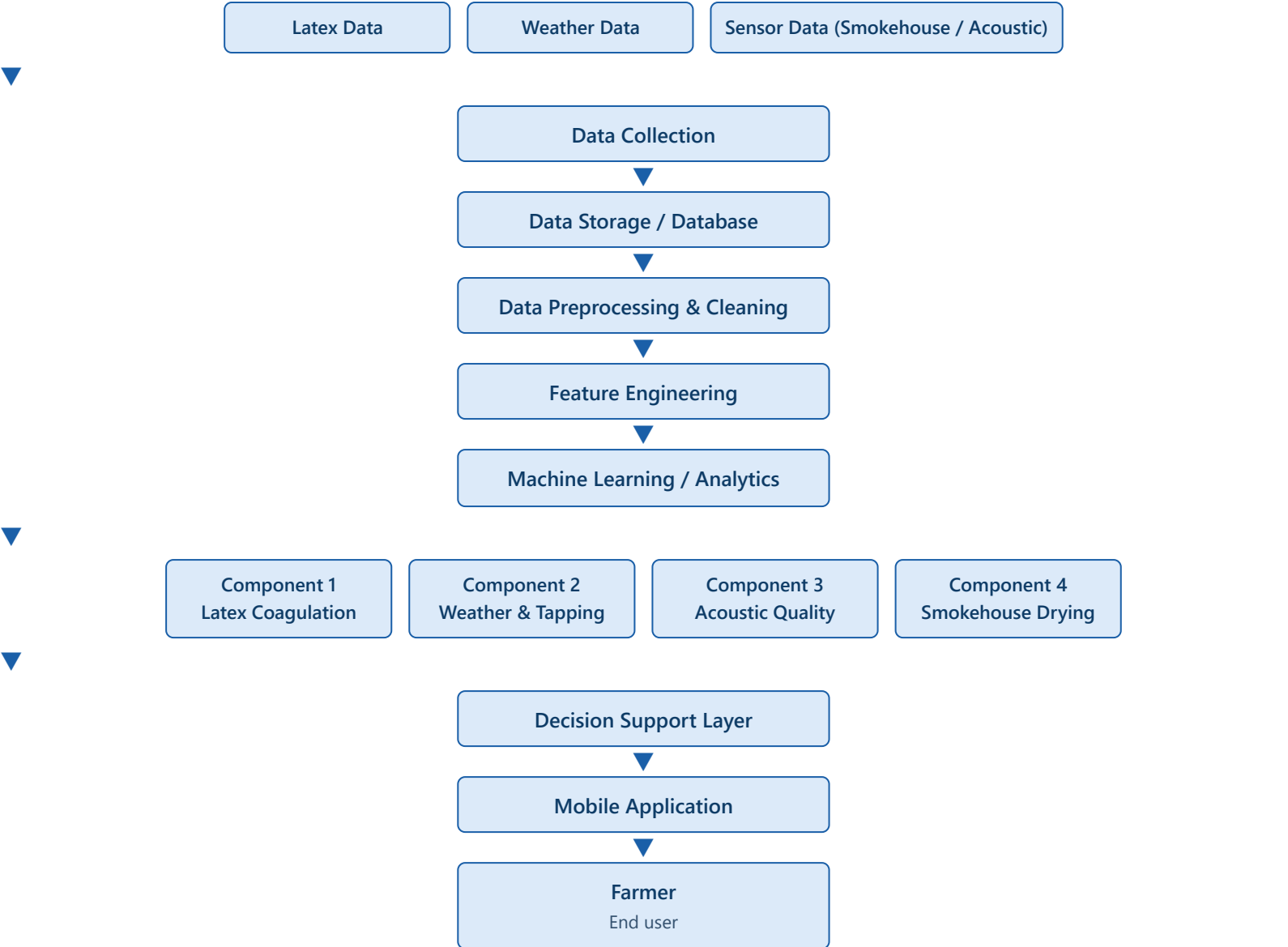


Figure 1. Overall architecture of the Intelligent Rubber Processing Framework, showing data sources, the shared data pipeline, the four research components, and delivery of decision support to the farmer through a mobile application.

2.3 Four Research Components Overview

Table 1. Overview of the four research components.

#	Component	Primary Focus	Key Output(s)
1	Smartphone-Based ML Decision Support for Latex Coagulation Management	Temperature-corrected DRC estimation and coagulation guidance	Corrected DRC, water & formic-acid dosage, coagulation-time prediction
2	Village-Level Weather & Tapping Advisory System	Weather-sensitive tapping and disease-risk advisory	Tap / Not-Tap recommendation, DRC stress level, bark-rot risk
3	Mobile Acoustic “Thud-Test” App for Rubber-Sheet Quality Assessment	Acoustic-signal-based sheet quality/moisture estimation	Acoustic features, quality/moisture prediction
4	Smart Smokehouse Drying-Time Prediction and Fire-Risk Monitoring System	IoT-based layer-wise drying prediction and safety monitoring (primary component)	Layer-wise remaining drying time, fire-risk alerts, farmer notifications

Scope note

Component 4 — the Smart Smokehouse Drying-Time Prediction and Fire-Risk Monitoring System — is the component assigned to this researcher and is therefore treated in substantially greater technical depth throughout this document (Chapter 6 onward), while Components 1–3 are described at a level sufficient to explain their role within the overall framework.

3 Component 1 — Latex Coagulation Decision Support

3.1 Purpose and Problem

Component 1, *“Smartphone-Based Machine Learning Decision Support System for Rubber Latex Coagulation Management,”* helps determine appropriate latex coagulation conditions using Metrolac readings, latex temperature, and latex volume. Rubber latex Dry Rubber Content (DRC) estimation is affected by temperature, so a raw Metrolac reading alone can be misleading. Traditional practice requires a manual Metrolac reading, a printed reference chart, manual calculation, temperature correction by hand, and interpretation by the farmer — a process that is slow and prone to human error. Component 1 proposes a smartphone-based machine-learning decision support system that performs this correction automatically.

3.2 Inputs and Outputs

Table 2. Component 1 inputs and outputs.

Category	Variable	Description
Input	Metrolac Reading	Raw hydrometer-style reading taken from the latex sample
Input	Latex Temperature (°C)	Temperature of the latex at the time of measurement
Input	Raw Latex Volume (L)	Volume of latex collected for the batch
Output	Corrected DRC	Temperature-corrected Dry Rubber Content estimate
Output	Water Requirement	Recommended water addition for target DRC
Output	Formic Acid Requirement	Recommended coagulant (formic acid) dosage
Output	Coagulation Time Prediction	Predicted time required for coagulation to complete

3.3 Process Workflow

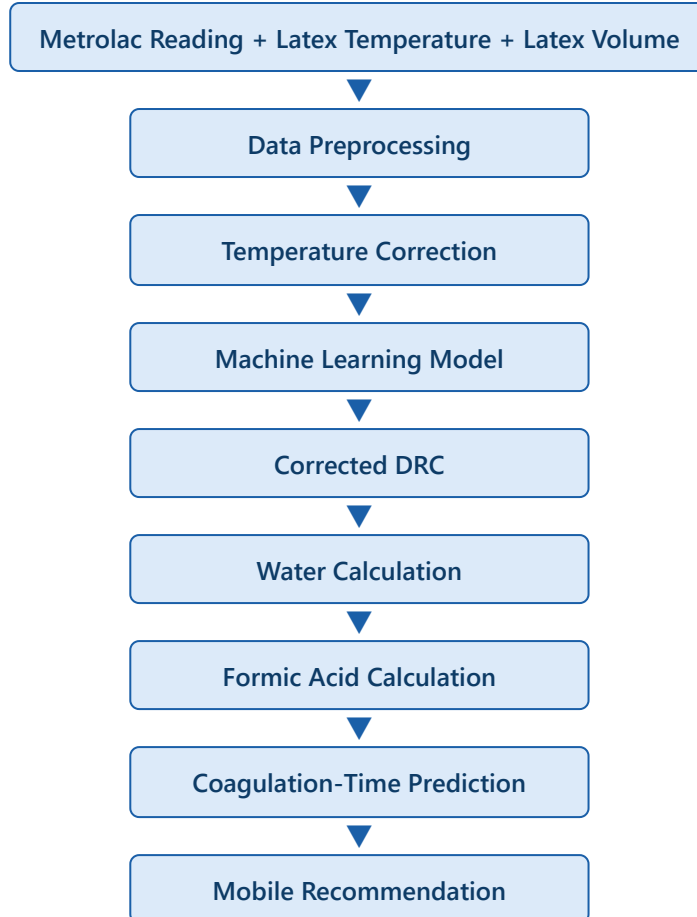


Figure 3. Component 1 process workflow, from raw field measurements to the mobile recommendation.

3.4 Predictive Model

The temperature-corrected linear-regression approach referenced for this component is denoted **Metro_N**. A previously referenced performance figure of **$R^2 = 0.84$** is associated with this approach in the source research material.

Distinguishing referenced results from new experimentation

The $R^2 = 0.84$ figure for Metro_N is carried forward from prior/available reference material associated with this research line, and is presented here for context only — it is **not** a result produced by this document. Final performance for this component must still be evaluated using the actual research dataset collected for this project, following the evaluation methodology set out in Section 9.2. No additional accuracy, RMSE, MAE, or other metric is claimed beyond this single referenced value.

4 Component 2 — Weather & Tapping Advisory System

4.1 Purpose

Component 2, the “*Village-Level Weather & Tapping Advisory System*,” provides weather-sensitive tapping recommendations for rubber farmers. It combines weather data, historical latex information, machine learning, and disease-risk prediction to generate an advisory that tells a farmer whether conditions are suitable for tapping on a given day.

4.2 Data Sources and Variables

Potential weather data sources include the OpenWeatherMap API, WeatherAPI, and local weather stations. The variables considered are rainfall (mm/day), relative humidity (%), temperature (°C), wet hours, consecutive rainy days, three-day cumulative rainfall, previous latex density, previous DRC, and tapping frequency.

4.3 Feature Engineering

- **Three-day cumulative rainfall** — recent wet-weather exposure, relevant to bark-rot risk.
- **Continuous wet hours** — duration of sustained wetness, relevant to tapping-panel condition.
- **Consecutive rainy days** — persistence of unfavourable weather.
- **Average humidity** — general moisture exposure of the tapping panel.
- **Previous latex density / previous DRC** — recent yield and quality trend.
- **Tapping frequency** — recent tapping load on the tree.

4.4 Machine Learning Models

Table 3. Component 2 features, predictions, and referenced machine-learning models.

Prediction	ML Model	Problem Type
Tap / Not Tap	Random Forest	Classification
DRC Stress Level	XGBoost	Classification / Prediction
Bark Rot Risk	Logistic Regression	Classification

These model choices are referenced as candidate/starting approaches for each prediction task; final model selection should be validated experimentally using the methodology in Chapter 9 once field data is available.

4.5 Workflow

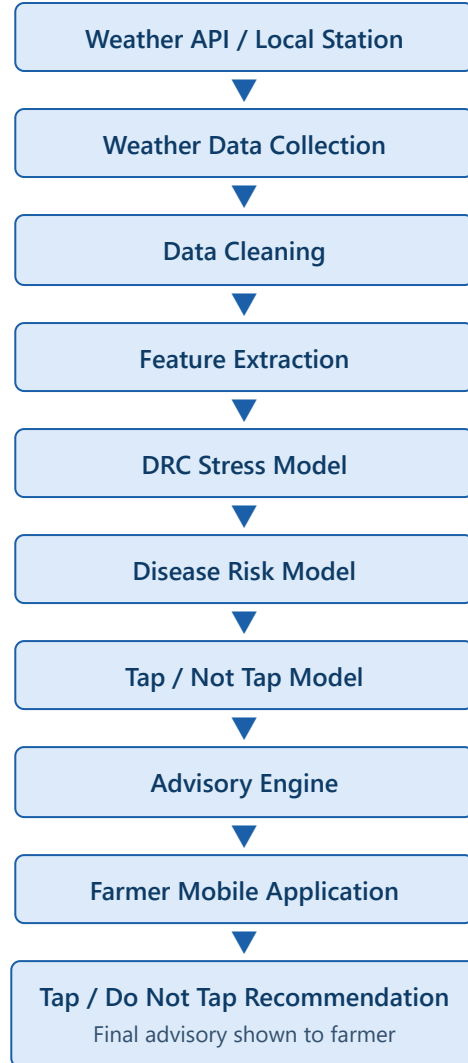


Figure 4. Component 2 weather and tapping advisory workflow.

5 Component 3 — Acoustic “Thud-Test” Quality Assessment

5.1 Purpose

Component 3, the *“Mobile Acoustic ‘Thud-Test’ App for Rubber Sheet Quality Assessment,”* uses smartphone acoustic signals to support rubber-sheet quality and moisture assessment. In operation, the farmer taps the rubber sheet, the smartphone microphone records the resulting sound, and the application processes the signal to estimate sheet quality.

5.2 Signal Processing Workflow

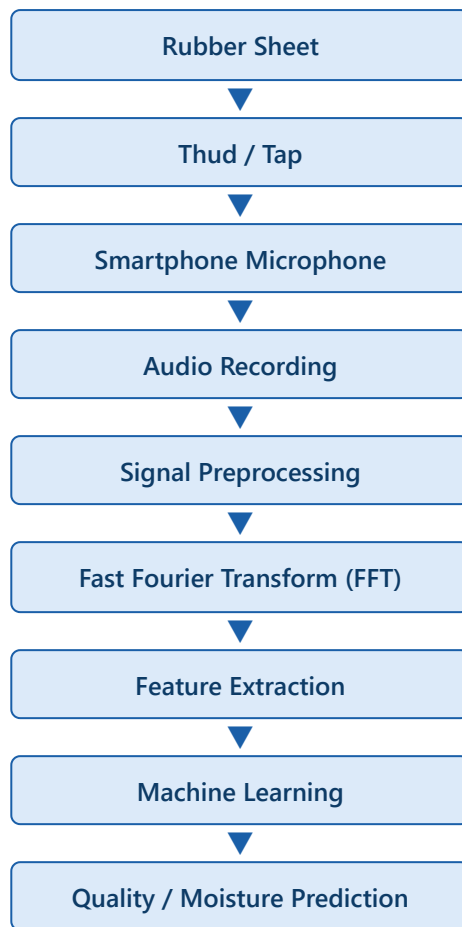


Figure 5. Component 3 acoustic signal-processing workflow, from the physical tap to the quality/moisture prediction.

5.3 Acoustic Features

Table 4. Component 3 acoustic features used for quality/moisture prediction.

Feature	Description
Dominant Frequency	The strongest frequency component of the recorded thud, related to sheet stiffness/density
Damping Ratio	Rate at which the acoustic signal decays, related to internal moisture and material consistency
Spectral Centroid	The “centre of mass” of the frequency spectrum, related to overall tonal quality

5.4 Predictive Model

The referenced model for this component is **k-Nearest Neighbours (kNN)**, with previously referenced performance figures of **RMSE = 0.660%** and **MAE = 0.480%**.

Distinguishing referenced results from new experimentation

As with Component 1, the RMSE and MAE figures above are carried forward from prior/available reference material and are presented for context only. They are **not** results generated by this document, and final performance for Component 3 must be evaluated using the project's own acoustic dataset once collected.

6 Component 4 — Smart Smokehouse Drying-Time Prediction and Fire-Risk Monitoring System

Primary research component

Component 4 is the component assigned to this researcher and is treated in the greatest technical depth in this document. It is an IoT-enabled intelligent smokehouse monitoring system.

6.1 Purpose and Objectives

Component 4, the “*Smart Smokehouse Drying-Time Prediction and Fire-Risk Monitoring System*,” develops an IoT-enabled intelligent smokehouse monitoring system with the following objectives:

- 1. Measure temperature and humidity in three layers of the smokehouse.
- 2. Use six environmental sensors in total (two per layer).
- 3. Collect sensor data every hour.
- 4. Record drying batches and the number of rubber sheets per batch.
- 5. Calculate sheet-loading density (sheets per square metre).
- 6. Predict remaining drying time separately for each layer.
- 7. Monitor abnormal heat conditions that may indicate fire risk.
- 8. Send relevant information and alerts to farmers through a mobile application.

6.2 Smokehouse Structure and Sensor Layout

The smokehouse is organised into three vertical layers, each fitted with one temperature sensor and one humidity sensor, giving six environmental sensors in total. A heat source (traditionally firewood) is positioned below the layers.

Table 5. Component 4 sensor layout across the three smokehouse layers.

Layer	Temperature Sensor	Humidity Sensor	Approximate Position
Layer 1	T1	RH1	Lowest layer, closest to the heat source
Layer 2	T2	RH2	Middle layer
Layer 3	T3	RH3	Highest layer, furthest from the heat source

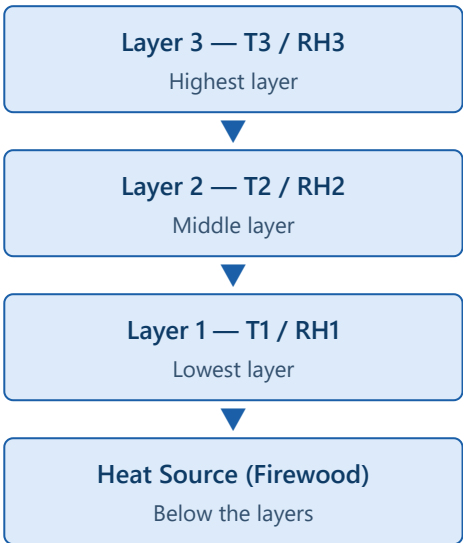


Figure 6. Component 4 smokehouse structure and six-sensor layout across three layers.

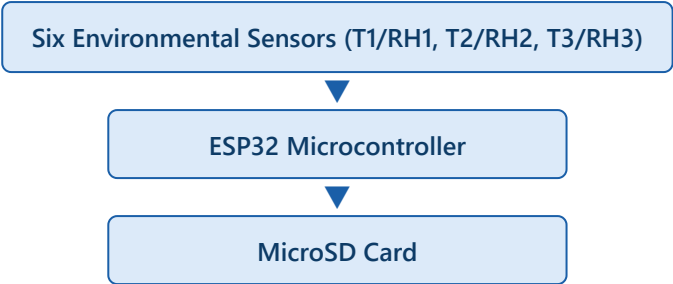
Research hypothesis, to be verified

The expected heat distribution between layers — typically warmer and drier nearer the heat source — is a research hypothesis based on general physical reasoning about heat and moisture transport. It must be verified through actual sensor measurements collected during this research, and should not be treated as an established finding until confirmed by field data.

6.3 IoT Device Architecture

The main microcontroller selected for the IoT device is the **ESP32**, chosen for its integrated Wi-Fi, low cost, and sufficient analogue/digital input capacity for six sensors. Two deployment configurations are considered.

Initial research prototype (offline)



Allows data collection without Wi-Fi connectivity, suitable for initial field trials.

Figure 8. Offline MicroSD-based data-logging architecture for early field prototyping.

Final online system

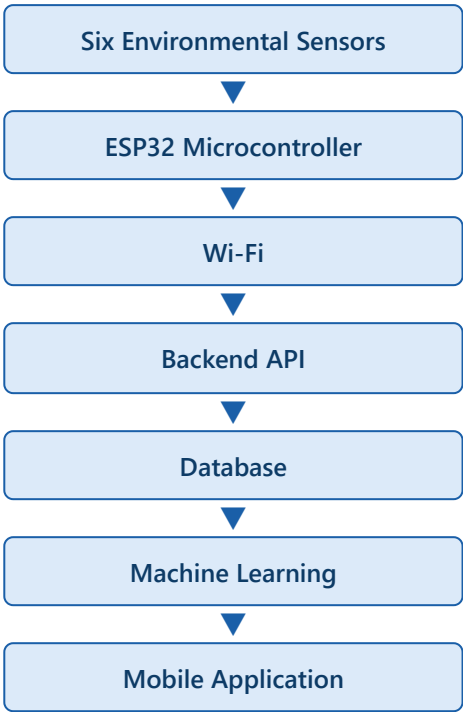


Figure 9. Online IoT → Backend → Machine Learning → Mobile Application architecture for the deployed system.

Figure 7. Six-sensor IoT architecture: two sensors (temperature and humidity) per smokehouse layer, connected to a single ESP32 controller.

6.4 Sensor Collection Frequency and Notification Schedule

Final decision: sensor data is collected every hour. Hourly collection was selected over higher-frequency alternatives because it provides higher temporal resolution than sparse manual checks, produces a better training dataset for the drying-time model, captures meaningful temperature and humidity changes, supports dynamic predictions that update as drying progresses, and supports safety monitoring without generating excessive data volume or notification noise.

Table 16. Component 4 data-collection and notification schedule.

Activity	Schedule
Sensor measurement	Every 1 hour
Database update	Every 1 hour
Machine-learning prediction	Every 1 hour
Normal farmer summary notification	Every 6 hours (optional/configurable)
Warning-level notification	Immediately
Critical-level notification	Immediately

6.5 Layer-Wise Drying-Time Prediction

The predictive model provides a separate remaining-drying-time prediction for each of the three layers, since each layer experiences different environmental conditions and therefore dries at a different rate.

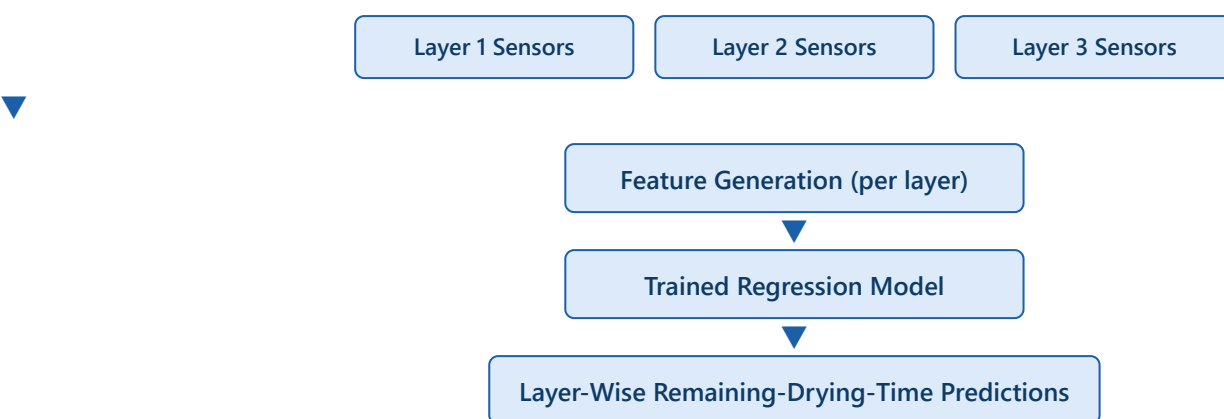


Figure 10. Layer-wise drying-time prediction workflow: each layer is scored independently by the same trained model using its own features.

Illustrative example only

L1 = 14.3 hours, L2 = 17.8 hours, L3 = 21.2 hours. **ILLUSTRATIVE / SYNTHETIC DATA — NOT AN EXPERIMENTAL RESULT** — these numbers illustrate the expected *shape* of the output (three independent predictions) and are not experimental results.

6.6 Sheet-Loading Density

The number of sheets loaded onto a layer, together with the layer's usable area, determines the sheet-loading density. Loading density may influence airflow, moisture removal, drying rate, and therefore drying time, and is included in the model as an explanatory feature.

Formula

$\text{Sheets_per_m}^2 = \text{Number_of_Sheets} \div \text{Layer_Area_m}^2$

Example: $180 \text{ sheets} \div 6 \text{ m}^2 = 30 \text{ sheets/m}^2$. **ILLUSTRATIVE / SYNTHETIC DATA — NOT AN EXPERIMENTAL RESULT**

6.7 Final Dataset Design

Table 6. Component 4 final dataset columns.

Group	Column	Description
Identifiers	Batch_ID	Unique identifier for one complete drying operation
Identifiers	Timestamp	Date and time of the hourly reading
Identifiers	Layer_ID	Which of the three layers the reading belongs to
Loading	Number_of_Sheets	Count of rubber sheets loaded on the layer for this batch
Loading	Layer_Area_m2	Usable area of the layer in square metres
Loading	Sheets_per_m2	Derived sheet-loading density (see Section 6.6)
Sensors	Temperature_C	Hourly temperature reading for the layer
Sensors	Humidity_%	Hourly relative-humidity reading for the layer
Progress	Elapsed_Drying_Hours	Hours elapsed since the batch started drying
Derived	Temperature_Change_1h	Change in temperature since the previous hourly reading
Derived	Humidity_Change_1h	Change in humidity since the previous hourly reading
Target	Remaining_Drying_Hours	Hours remaining until the layer is fully dried (prediction target)
Safety	Fire_Risk_Level	Rule-based safety classification: Normal / Warning / Critical

6.8 Removed Parameters

Table 7. Parameters deliberately excluded from the predictive feature set, and the reasons why.

Parameter	Reason for Exclusion
Firewood_Amount_kg	Impractical to measure accurately during actual smokehouse operation; excluded from the dataset.
Smokehouse_ID (as an ML feature)	The model should learn physical/environmental relationships rather than memorise smokehouse identity; may optionally be retained in administrative/raw records for traceability and generalisation analysis, but must not be used as a predictive input.

6.9 Batch Definition

A **Batch** is defined as one complete rubber-sheet drying operation. For example, Batch B001 consists of Layer 1, Layer 2, and Layer 3, each with hourly sensor records logged until that layer's drying is complete.

Table 8. Batch definition and related terms.

Term	Definition
Batch	One complete drying operation, identified by a unique Batch_ID
Batch record	The set of all hourly, per-layer readings belonging to one batch
Batch completion	The point at which every layer in the batch has finished drying

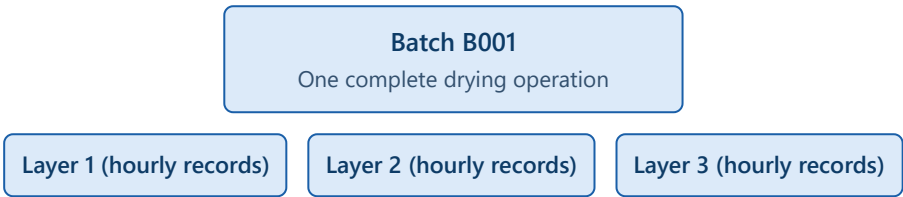


Figure 19. Batch hierarchy: one batch contains three layers, each with its own hourly record sequence.

6.10 Target Variable and Data-Leakage Warning

The prediction target is **Remaining_Drying_Hours**, computed during dataset construction as:

Formula

Remaining_Drying_Hours = Actual_Total_Drying_Hours – Elapsed_Drying_Hours

Example: if the actual drying duration for a layer is 20 hours and 5 hours have elapsed, the remaining time is 15 hours.

ILLUSTRATIVE / SYNTHETIC DATA — NOT AN EXPERIMENTAL RESULT

Critical data-leakage warning

Actual_Total_Drying_Hours is used only to construct the target variable during dataset preparation. It must **never** be used as an input feature at prediction time, because it would not be known in advance during real-time operation and its inclusion as a feature would leak the answer into the model, invalidating the evaluation.

6.11 Different Drying Time by Layer

Each layer can have a different actual total drying duration because of its position relative to the heat source and airflow. Illustrative values — Layer 1 = 20 hours, Layer 2 = 23 hours, Layer 3 = 27 hours — are examples only; actual values must be collected from field experiments.

ILLUSTRATIVE / SYNTHETIC DATA — NOT AN EXPERIMENTAL RESULT

6.12 Machine Learning Models for Drying-Time Prediction

Remaining-drying-time prediction is framed as a regression problem with target **Remaining_Drying_Hours**. Five candidate models are considered:

Table 9. Candidate machine-learning models for layer-wise drying-time prediction.

Model	Type	Purpose
Linear Regression	Regression	Baseline model
Random Forest Regressor	Regression	Captures nonlinear relationships
Gradient Boosting Regressor	Regression	Strong general-purpose tabular baseline
XGBoost Regressor	Regression	Advanced gradient boosting
Extra Trees Regressor	Regression	Ensemble comparison model

No model is declared best in advance; the best-performing model must be selected using the actual evaluation results obtained in Section 9.2, once real field data has been collected.

6.13 Feature Engineering

Core engineered features and their formulas.

Feature	Formula / Definition
Sheets_per_m2	Number_of_Sheets ÷ Layer_Area_m2
Temperature_Change_1h	Current temperature – previous-hour temperature
Humidity_Change_1h	Current humidity – previous-hour humidity
Elapsed_Drying_Hours	Current timestamp – batch start timestamp

Potential future features include a rolling temperature mean, rolling humidity mean, temperature trend, and humidity trend. As a general rule, only features that are available at prediction time may be used — consistent with the data-leakage warning in

6.14 Fire-Risk Monitoring System

The initial approach to fire-risk monitoring is **rule-based**, using three risk levels.

Table 11. Fire-risk levels used by the rule-based monitoring system.

Level	Meaning
NORMAL	Environmental readings are within the expected range for the current drying stage
WARNING	Readings suggest an abnormal pattern that warrants farmer inspection
CRITICAL	Readings suggest an immediate, high-priority safety concern

Potential indicators include current temperature, the rate of temperature increase, temperature change, humidity, the specific layer, and the drying stage. High temperature alone does not necessarily indicate fire; the system is intended to identify abnormal conditions or patterns rather than react to a single absolute value.

Thresholds are not invented

The exact numerical thresholds separating Normal, Warning, and Critical must be established using field observations, relevant literature, expert/industry knowledge, safety requirements, and experimental data gathered during this research. No specific threshold values are asserted in this document.

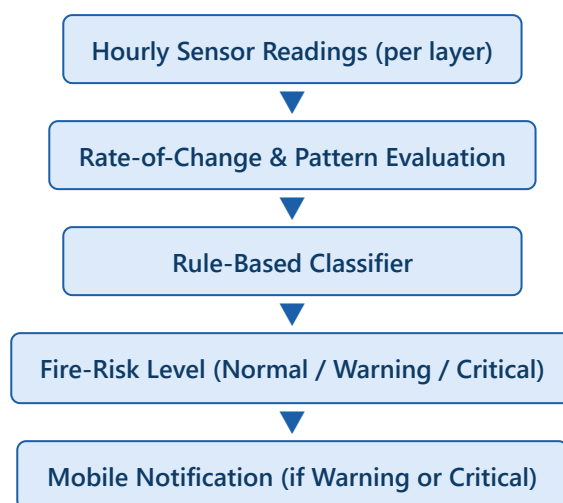


Figure 11. Fire-risk monitoring workflow.

6.15 Optional Fire-Risk Machine-Learning Models

If sufficient labelled fire-risk data becomes available during the research, the following classifiers may be investigated as a possible enhancement to the rule-based approach:

Table 12. Optional fire-risk machine-learning models to be investigated if labelled data permits.

Model	Type
Logistic Regression	Classification
Decision Tree	Classification
Random Forest	Classification
XGBoost Classifier	Classification
Support Vector Machine (SVM)	Classification

Evaluation of any such classifier should include accuracy, precision, recall, F1-score, and a confusion matrix. **Recall is particularly important** for this safety-critical use case, because a missed critical event (a false negative) is far more costly than a false alarm (a false positive) — the monitoring system should be biased toward catching true risk events even at the cost of some extra warnings.

7 Correlation Analysis

A dedicated correlation-analysis section is included at the supervisor's request. The analysis investigates relationships among the core Component 4 variables: Number_of_Sheets, Layer_Area_m2, Sheets_per_m2, Temperature_C, Humidity_%, Elapsed_Drying_Hours, Temperature_Change_1h, Humidity_Change_1h, and Remaining_Drying_Hours.

7.1 Variables Considered

Table 13. Variables included in the correlation analysis.

Variable	Role
Number_of_Sheets	Loading variable
Layer_Area_m2	Loading variable
Sheets_per_m2	Derived loading-density variable
Temperature_C	Sensor variable
Humidity_%	Sensor variable
Elapsed_Drying_Hours	Progress variable
Temperature_Change_1h	Derived sensor-trend variable
Humidity_Change_1h	Derived sensor-trend variable
Remaining_Drying_Hours	Target variable

7.2 Pearson and Spearman Correlation

Two complementary correlation measures are planned. **Pearson correlation** measures the strength of a linear relationship between two variables. **Spearman correlation** measures monotonic relationships and is less dependent on strict linearity, making it useful when a relationship is consistent in direction but not perfectly linear. A correlation coefficient of +1 indicates a strong positive relationship, 0 indicates no linear relationship, and -1 indicates a strong negative relationship. The analysis will also include feature-versus-target ranking and a multicollinearity check across predictor variables.

7.3 Correlation Heatmap

Figure 12 shows an illustrative correlation heatmap in the intended layout and colour scheme (blue for positive correlation, red for negative correlation). The specific coefficient values shown are synthetic placeholders and must be replaced with values computed from the actual research dataset once it is collected.

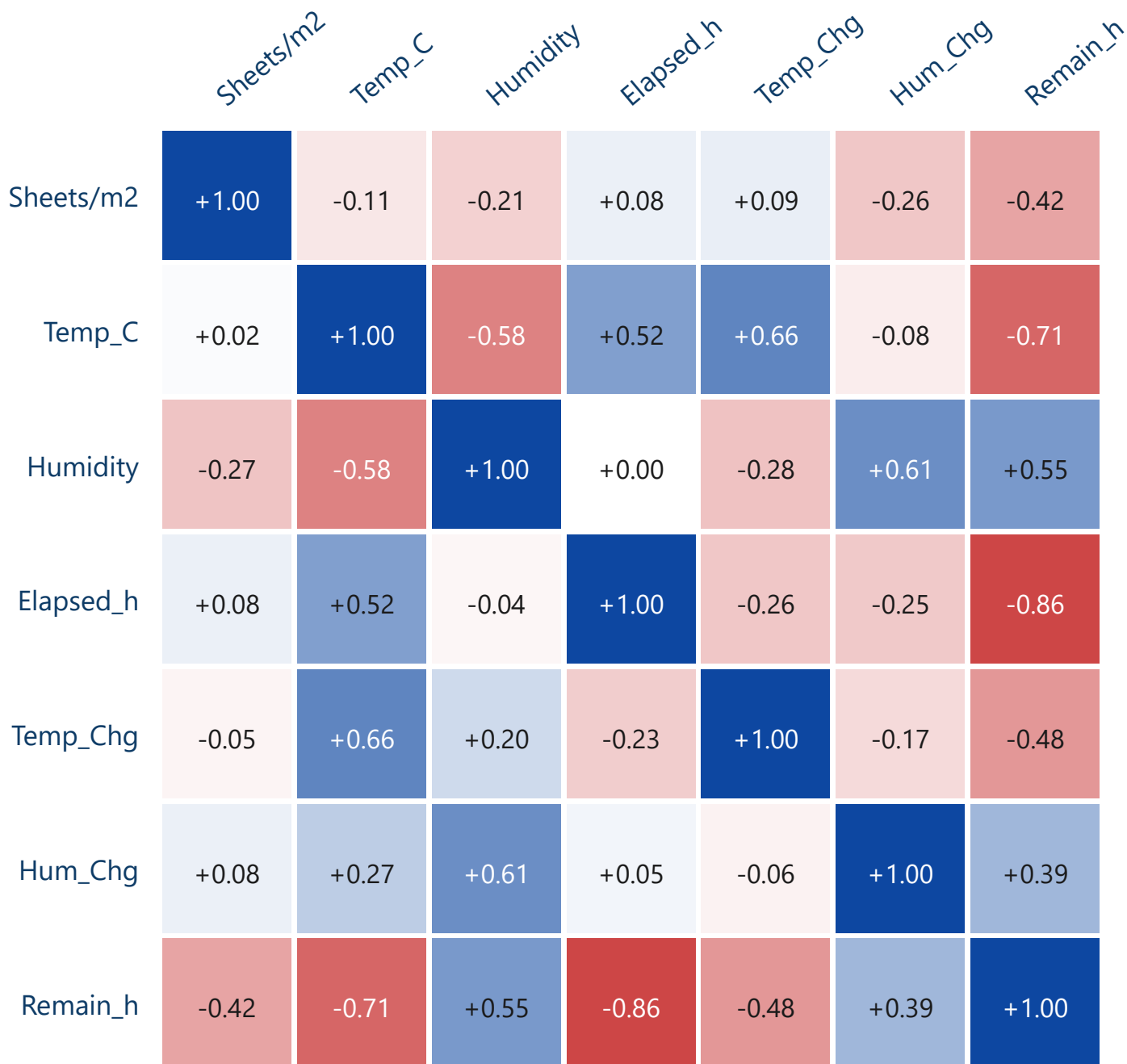


Figure 12. Illustrative correlation heatmap for the Component 4 variables (synthetic values).

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7.4 Interpretation Guidelines

General correlation-strength interpretation guidelines.

r Range	Interpretation
0.00 – 0.29	Weak
0.30 – 0.69	Moderate
0.70 – 1.00	Strong

These bands are general interpretation guidelines, not fixed rules; a feature should not be removed from the model solely because of its correlation coefficient. A special case applies to **Sheets per m2**, which is mathematically derived from

Number_of_Sheets ÷ Layer_Area_m2 — correlations involving these three variables must therefore be interpreted carefully, since part of any observed relationship is a direct mathematical consequence of the derivation rather than new information.

8 Data Preprocessing, Quality and Splitting

8.1 Preprocessing Pipeline

Table 14. Component 4 data-preprocessing pipeline.

Step	Action
1	Load raw data
2	Validate timestamps
3	Check missing values
4	Check duplicate records
5	Check sensor failures
6	Detect impossible values
7	Validate Layer_ID
8	Validate Batch_ID
9	Validate sheet counts
10	Validate layer area
11	Calculate Sheets_per_m2
12	Calculate elapsed time
13	Calculate temperature change
14	Calculate humidity change
15	Generate Remaining_Drying_Hours
16	Check target values
17	Prepare model dataset

8.2 Data Quality Checks

- Missing temperature or humidity readings
- Duplicate timestamps
- Sensor disconnection
- Flat-line (stuck) readings
- Timestamp errors
- Negative elapsed time
- Incorrect Batch_ID values
- Incorrect Layer_ID values
- Invalid sheet counts
- Invalid layer areas
- Negative remaining drying time
- Incorrect drying-completion records

8.3 Batch-Aware Data Splitting

Important: do not split randomly by row

Hourly records from the same batch are highly related to one another, so randomly splitting individual rows between training and test sets can leak information from a batch's training portion into its own test portion. Data must instead be split at the **Batch_ID** level, so that all records from a given batch fall entirely within one split.

Example Batch_ID-based train/validation/test split (illustrative range only).

Split	Example Batch Range
Training	B001 – B080
Validation	B081 – B090
Testing	B091 – B100

These specific batch numbers are examples only and are used purely to illustrate the splitting strategy; the real split boundaries will depend on the number of batches actually collected.

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9.1 Overall Data Science Pipeline

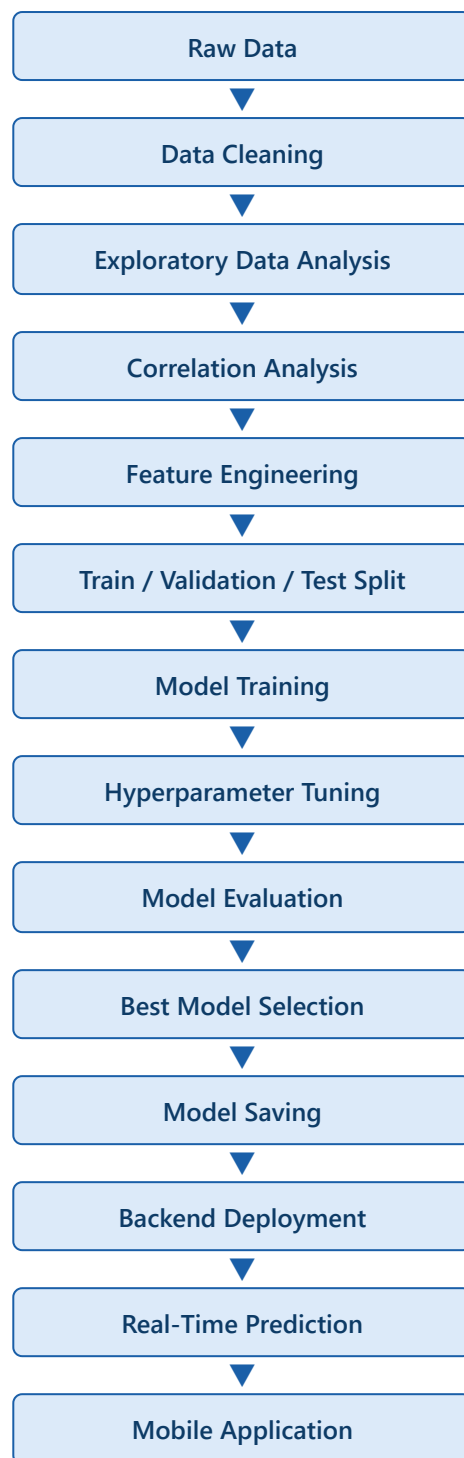


Figure 13. Overall data-science / machine-learning training pipeline, from raw data to the mobile application.

9.2 Regression Evaluation Metrics

Table 10. Regression evaluation metrics used for the drying-time models.

Metric	Meaning	Better Direction
MAE	Mean Absolute Error — average absolute prediction error	Lower
RMSE	Root Mean Squared Error — penalises large errors more heavily	Lower
R ²	Coefficient of determination — proportion of variance explained	Higher

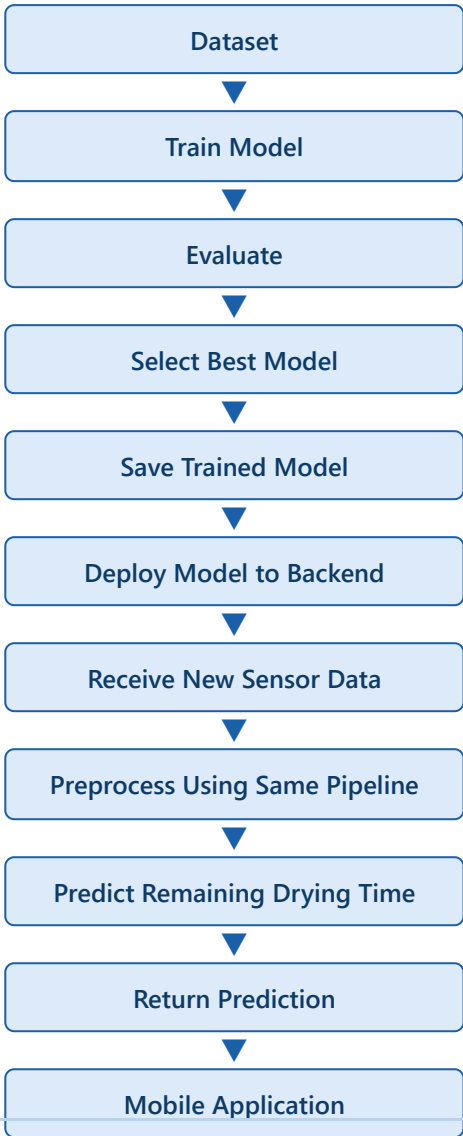
Model comparison table — to be completed with actual results once experimentation is performed.

Model	MAE	RMSE	R ²	Rank
Linear Regression	TBD	TBD	TBD	TBD
Random Forest	TBD	TBD	TBD	TBD
Gradient Boosting	TBD	TBD	TBD	TBD
XGBoost	TBD	TBD	TBD	TBD
Extra Trees	TBD	TBD	TBD	TBD

No values are invented

Every cell in the model-comparison table above is marked “TBD” (to be determined through experimentation). No accuracy, RMSE, MAE, R², or other performance value is invented for Component 4; these cells will be completed only once real experiments have been run on field-collected data.

9.3 Model Deployment Workflow



9.4 Real-Time Prediction Example

At, for example, 10:00, sensors measure current temperature and humidity for Layer 1, Layer 2, and Layer 3. The backend receives this data, generates the engineered features described in Section 6.13, and the trained model predicts a remaining drying time for each layer independently. The fire-risk rules of Section 6.14 are evaluated for all three layers, and the mobile application updates accordingly. At 11:00, the entire cycle repeats using the newly collected hour of data.

9.5 Model Monitoring

After deployment, the system should be monitored for prediction error, sensor data quality, missing readings, model drift, unusual environmental conditions, false warnings, missed warnings, and farmer feedback, so that the model and rule thresholds can be refined over time as more field data becomes available.

10 System Architecture — IoT, Backend and Mobile Application

10.1 Complete IoT Architecture

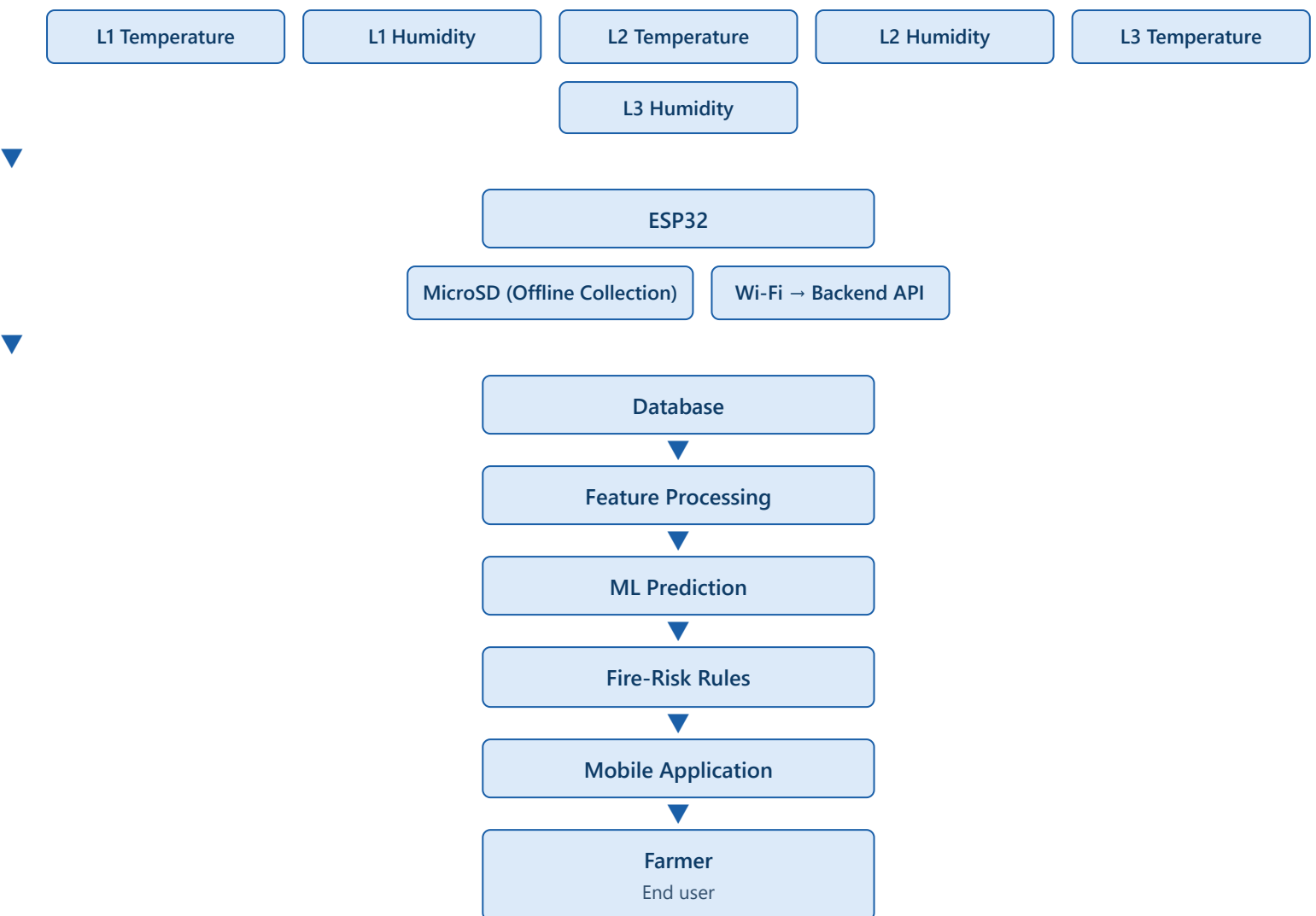


Figure 20. Complete IoT architecture connecting the six smokehouse sensors through the ESP32, backend, database, machine-learning service, fire-risk rules, and mobile application to the farmer.

10.2 Backend Responsibilities

The ESP32 device sends hourly sensor readings to the backend. The backend receives the sensor data, validates it, stores it in the database, generates the engineered features, calls the trained machine-learning model for prediction, stores the prediction, evaluates the fire-risk rules, and sends the combined results to the mobile application.

10.3 Mobile Application and Dashboard

The mobile application displays, per batch and layer: Batch ID, layer, temperature, humidity, remaining drying time, fire-risk status, historical trends, and notifications.

Table 17. Mobile application information sources.

Information Shown	Source
Batch ID and layer	Backend database
Current temperature / humidity	Latest hourly sensor reading
Remaining drying time	Machine-learning prediction service
Fire-risk status	Rule-based (and optionally ML-based) fire-risk service
Historical trends	Stored time-series of past readings and predictions
Notifications	Generated per the schedule in Table 16

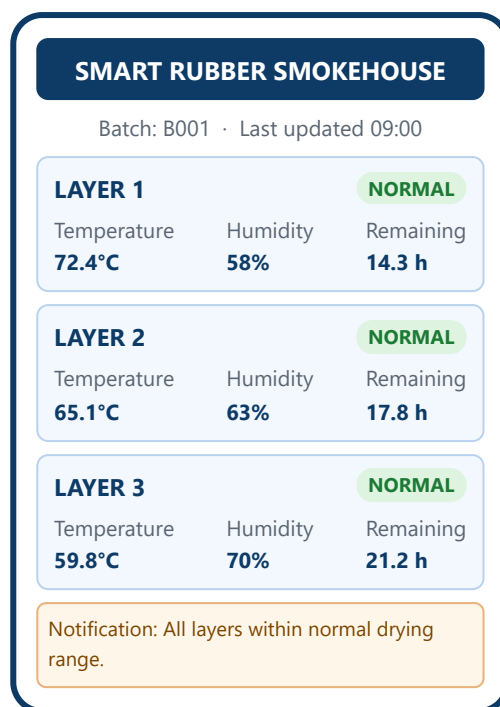


Figure 18. Illustrative mobile dashboard mock-up for the Smart Smokehouse application.

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10.4 Notification System

Notifications follow the schedule defined in Table 16: routine sensor and prediction updates every hour, an optional normal-condition summary every six hours, and immediate notifications for both Warning and Critical fire-risk levels. An example Warning-level message reads: “*Layer 1 temperature is increasing rapidly. Please inspect the layer and heat source.*” Such an alert reports an abnormal pattern; it does not, by itself, prove that an actual fire is occurring.

11 Sample Datasets (Illustrative)

All data in this chapter is synthetic

Every dataset column list, sample row, and chart in this chapter is provided to illustrate the intended data structure, dataset design, and user-interface behaviour. **ILLUSTRATIVE / SYNTHETIC DATA — NOT AN EXPERIMENTAL RESULT** None of the values below should be interpreted as an actual research finding.

11.1 Component 1–3 Sample Data Columns

Illustrative sample dataset columns for Components 1–3.

Component	Sample Columns
Component 1	Metrolac_Reading, Latex_Temperature, Latex_Volume, Corrected_DRC, Water_Required, Formic_Acid_Required, Coagulation_Time
Component 2	Rainfall, Temperature, Humidity, Wet_Hours, Previous_DRC, Previous_Density, Tapping_Frequency, Tap_Decision, DRC_Stress, Bark_Rot_Risk
Component 3	Dominant_Frequency, Damping_Ratio, Spectral_Centroid, Moisture/Quality_Target

11.2 Component 4 Sample Dataset and Records

Component 4's sample dataset columns are listed in Table 6 (Section 6.7). Illustrative hourly rows for Batch B001, Layer 1 are shown below.

Illustrative sample hourly records for Batch B001, Layer 1.

Batch	Time	Layer	Sheets	Area (m²)	Sheets/m²	Temp (°C)	Hum (%)	Elapsed (h)	ΔTemp	ΔHum	Remaining (h)	Risk
B001	06:00	L1	180	6.0	30.0	68.2	64	0	—	—	20	Normal
B001	07:00	L1	180	6.0	30.0	69.4	63	1	+1.2	−1	19	Normal
B001	08:00	L1	180	6.0	30.0	71.0	61	2	+1.6	−2	18	Normal
B001	09:00	L1	180	6.0	30.0	73.5	59	3	+2.5	−2	17	Warning

Illustrative sample hourly records for Layer 2 and Layer 3 of the same batch.

Batch	Time	Layer	Temp (°C)	Hum (%)	Elapsed (h)	Remaining (h)	Risk
B001	06:00	L2	65.0	69	0	23	Normal
B001	07:00	L2	65.9	68	1	22	Normal
B001	06:00	L3	59.1	74	0	27	Normal
B001	07:00	L3	59.6	73	1	26	Normal

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11.3 Illustrative Charts

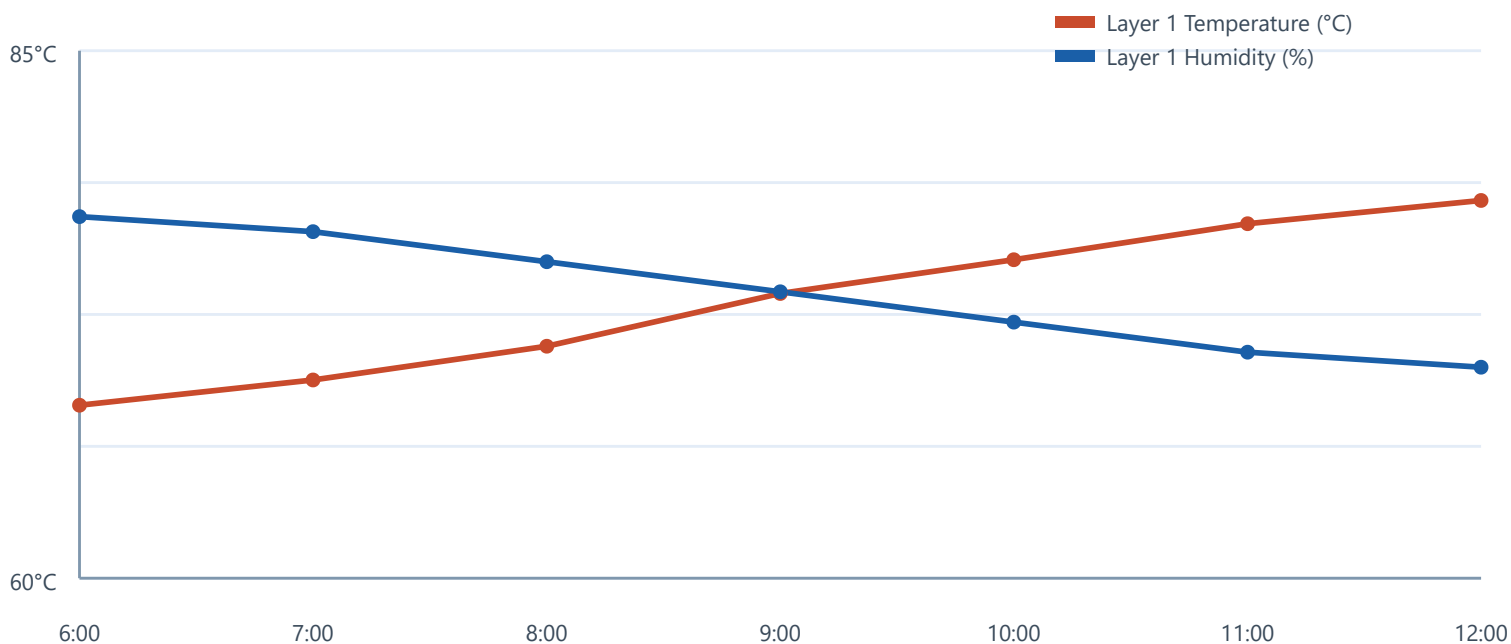


Figure 16. Illustrative Component 4 Layer 1 temperature and humidity trend over a drying session (synthetic data).

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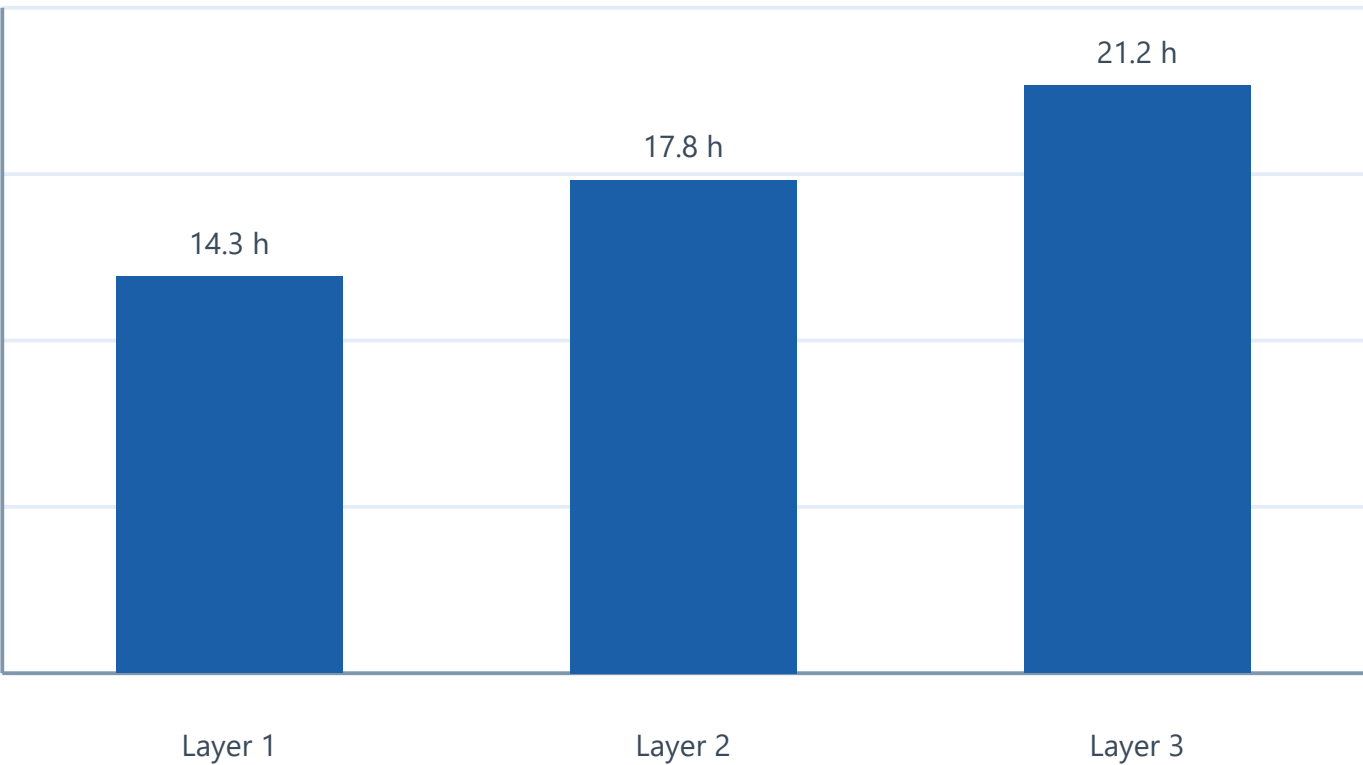


Figure 17. Illustrative remaining drying-time comparison across the three layers at a given point in time (synthetic data).

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12 Research Data Collection Plan and Field Ethics

12.1 Data Collection Plan by Component

- **Component 1** — collect latex measurements (Metrolac reading, temperature, volume) alongside laboratory or reference DRC values where available.
- **Component 2** — collect weather data and corresponding tapping/yield information.
- **Component 3** — collect acoustic recordings of rubber-sheet taps alongside quality/moisture reference measurements.
- **Component 4** — collect hourly smokehouse sensor data across multiple batches and, where possible, multiple smokehouses.

The Component 4 field data collection procedure is as follows:

1. Obtain permission from the smokehouse owner/operator and relevant research authorities.
2. Visit the smokehouse and identify the three layers.
3. Install the six sensors (temperature and humidity per layer).
4. Verify that all sensor readings are plausible before starting collection.
5. Record the layer area for each layer.
6. Count the number of sheets loaded per layer.
7. Create a unique Batch_ID for the drying operation.
8. Start hourly data collection.
9. Continue collection until drying is complete for all layers.
10. Record the actual completion time for each layer.
11. Repeat this process across many batches and, where possible, multiple smokehouses.
12. Assemble the final research dataset from the collected batches.

12.2 Field Data Collection Ethics and Permission

Ethics and permission

Permission should be obtained from the relevant smokehouse owner/operator and appropriate university/research authorities before any sensor installation or data collection begins. Data collection should minimise disruption to normal processing activity, and any sensitive operational information obtained during fieldwork should be handled appropriately and confidentially.

13 Research Evaluation Plan

The overall framework, and Component 4 in particular, will be evaluated across four complementary dimensions.

Research evaluation dimensions and associated evaluation aspects.

Dimension	Evaluation Aspects
Technical performance	Sensor reliability, data completeness, communication reliability, backend response time, prediction latency
Machine-learning performance	MAE, RMSE, R^2 for drying-time regression; accuracy, precision, recall, F1-score, and confusion matrix if a fire-risk classifier is used
Application performance	Usability, notification effectiveness, farmer understanding of the displayed information
Field performance	Drying-time prediction accuracy, layer-wise accuracy, and practical usefulness reported by farmers and processors

14 Research Novelty and Final Contribution

The research contribution is the **integrated framework** that connects rubber latex decision support, weather/tapping intelligence, acoustic rubber-sheet assessment, IoT smokehouse monitoring, layer-wise drying-time prediction, fire-risk monitoring, and mobile decision support into a single, coherent system. Novelty is described as the proposed integration and application of these techniques to the smallholder rubber-processing context, rather than a claim that any individual underlying technique (machine learning, IoT sensing, acoustic analysis) is itself new.

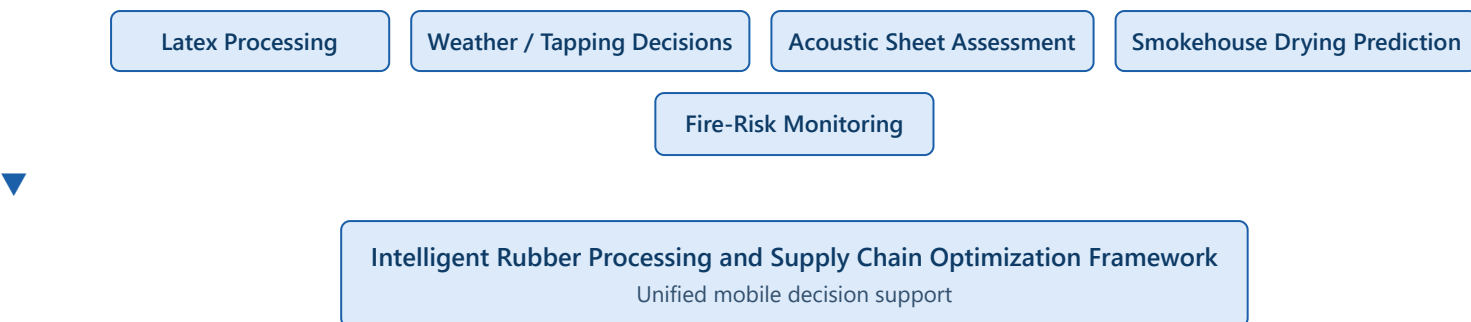


Figure 21. Convergence of the four research components into the unified Intelligent Rubber Processing and Supply Chain Optimization Framework.

15 Expected Benefits, Limitations and Future Work

15.1 Expected Benefits

The following benefits are *expected* outcomes of the proposed framework; they are not confirmed results.

Table 20. Expected benefits of the proposed intelligent framework.

#	Expected Benefit
1	Better decision support for rubber-processing activities
2	Reduced dependence on fixed, one-size-fits-all drying times
3	Improved understanding of layer-wise drying behaviour
4	Real-time smokehouse monitoring
5	Early warning of abnormal heat/fire-risk conditions
6	Better use of environmental and weather data
7	Improved rubber-processing efficiency
8	Potential improvement in product quality
9	Reduced unnecessary manual monitoring effort
10	Mobile access to processing information for farmers

15.2 Limitations

Table 19. Potential research limitations.

#	Potential Limitation
1	Sensor accuracy under harsh smokehouse conditions
2	Smokehouse environmental harshness (heat, smoke, humidity) affecting hardware longevity
3	Limited initial dataset size during early field trials
4	Variation between different smokehouse designs
5	Limited number of labelled fire-risk events for ML-based fire-risk modelling
6	Sensor failures or disconnection
7	Limited or unreliable internet availability at some sites
8	Generalisation of trained models between different smokehouses
9	Farmer adoption and willingness to use the mobile application
10	Difficulty obtaining precise drying-completion labels in the field
11	Potential distribution shift between training conditions and future operating conditions

15.3 Future Work

- Additional sensors and environmental variables (e.g. smoke density, airflow).
- Automated smokehouse control (ventilation and heat).
- Advanced time-series models, including LSTM/GRU and other temporal architectures.
- Edge AI / TinyML deployment directly on the ESP32 or similar devices.

- Automated ventilation and heat control informed by model predictions.
- A larger, multi-region dataset spanning many smokehouses.
- More sophisticated, data-driven fire-risk modelling.
- Cloud scalability for larger deployments.
- An offline-first mobile application for areas with unreliable connectivity.

16 Implementation Roadmap and Overall Research Workflow

16.1 Implementation Roadmap

Table 18. Sixteen-phase research implementation roadmap.

Phase	Activity
1	Literature Review
2	Problem Definition
3	System Design
4	Component Development
5	Data Collection
6	Data Preprocessing
7	Correlation & Exploratory Data Analysis
8	Machine-Learning Training
9	Model Evaluation
10	IoT Integration
11	Backend Development
12	Mobile Application Development
13	System Integration
14	Field Testing
15	Final Evaluation
16	Research Documentation

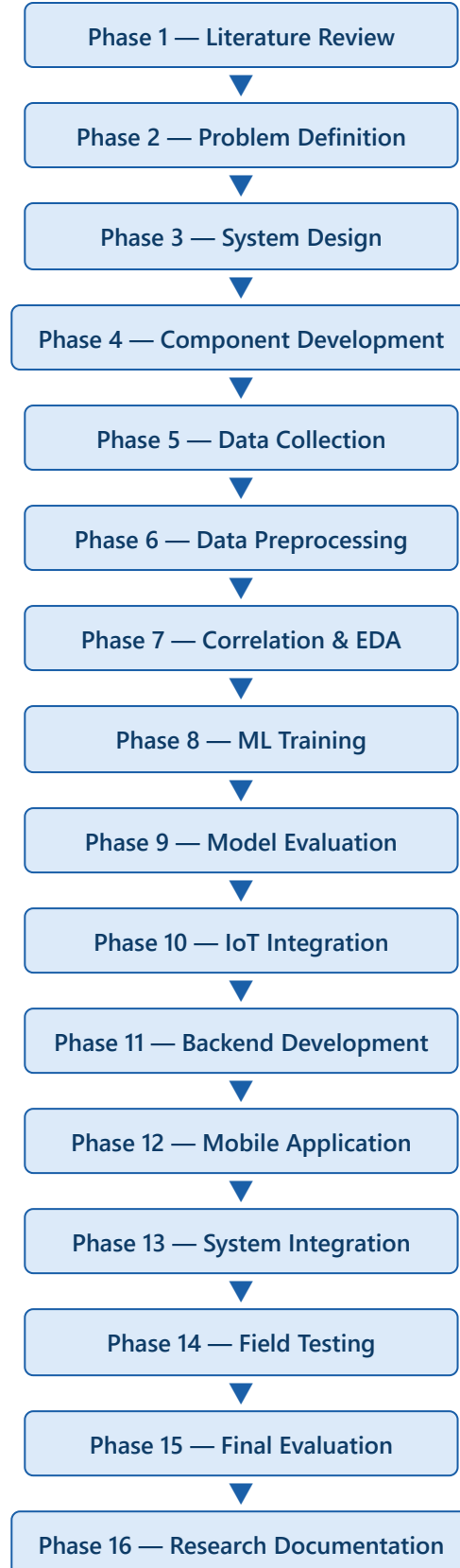
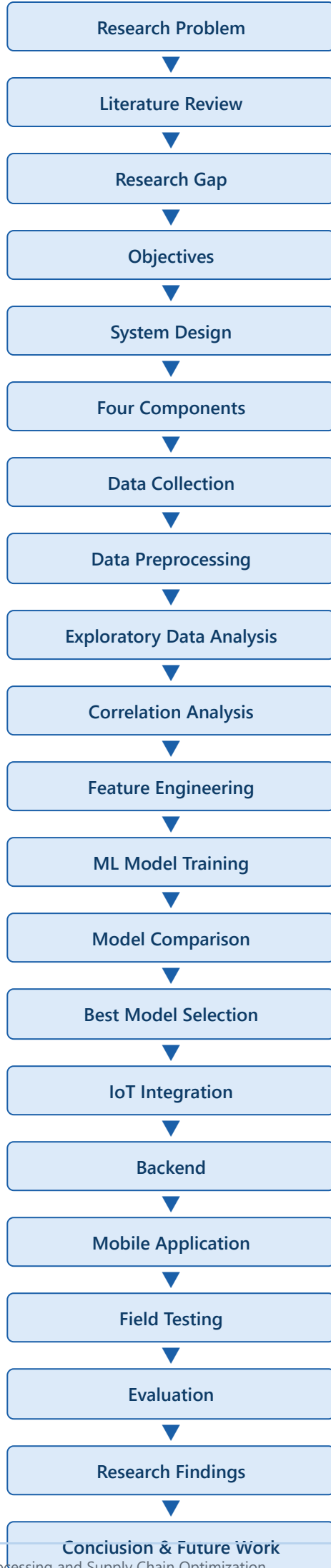


Figure 15. Visual research implementation roadmap across sixteen phases.

16.2 Overall Research Workflow



17 Technology Stack

Table 15 summarises the technology categories used across the framework. Specific product/library choices within each category may be refined during implementation.

Table 15. Technology stack summary across the framework.

Layer	Representative Technologies
Microcontroller / IoT hardware	ESP32, temperature sensors, humidity sensors, MicroSD module
Connectivity	Wi-Fi (online mode), MicroSD (offline mode)
Backend / API	REST-style backend API for data ingestion, validation and prediction serving
Database	Relational or time-series database for sensor, batch, and prediction records
Machine learning	Python-based ML stack (e.g. scikit-learn, XGBoost) for regression and classification models
Weather data	OpenWeatherMap API, WeatherAPI, or local weather station feeds
Signal processing	Digital signal processing / FFT libraries for acoustic feature extraction
Mobile application	Cross-platform or native mobile application framework
Notifications	Push-notification service integrated with the mobile application

18 Final Research Summary

This research proposes an **Intelligent Framework for Precision Rubber Processing and Supply Chain Optimization**, using machine learning, IoT, mobile computing, weather intelligence, and acoustic signal processing. The framework consists of four complementary components: latex-coagulation decision support (Component 1); weather and tapping advisory (Component 2); acoustic rubber-sheet quality assessment (Component 3); and smart smokehouse drying-time prediction and fire-risk monitoring (Component 4).

Component 4, the primary focus of this document, uses six sensors distributed across three smokehouse layers, collects environmental data hourly, incorporates sheet-loading density and drying progress as features, predicts remaining drying time separately for each layer, monitors for abnormal heat conditions, and provides information to farmers through a mobile application.

Taken together, the complete framework aims to transform selected rubber-processing activities into data-driven, intelligent decision-support services, while remaining explicit throughout about which figures are established reference values, which are illustrative/synthetic examples, and which are still to be determined through experimentation.

Experimental values must not be invented

All sample datasets, graphs, and prediction values presented in this document are provided for system demonstration, dataset design, workflow explanation, and user-interface demonstration. They are **not** final research results. Final results must be generated from actual field data collected and analysed during the course of this research.